Case I: $\angle A$ is acute.

Produce BO to meet the circle at D.

$\angle BDC = \angle BAC = A$

$\angle BCD = 90^\circ$

$\sin \angle BDC = \dfrac{BC}{BD}$ or $\sin A = \dfrac{a}{2R} \Rightarrow \dfrac{a}{\sin A} = 2R$

Case II: $\angle A$ is right angle.

In this case O must be on the side BC of the$\triangle ABC$.

Now, $\dfrac{a}{\sin A} = \dfrac{BC}{\sin 90^\circ} = \dfrac{2R}{1} = 2R \Rightarrow \dfrac{a}{\sin A} = 2R$

Case III: $\angle A$ is obtuse .

Produce BO to meet the circle at D.

$\angle BDC + \angle BAC = 180^\circ$

$\angle BDC = 180^\circ - \angle BAC = 180^\circ - A$

$\angle BCD = 90^\circ$

$\sin \angle BDC = \dfrac{BC}{BD}$ or $\sin(180^\circ - A) = \sin A = \dfrac{a}{2R} \Rightarrow \dfrac{a}{\sin A} = 2R$

In each case, we have $\dfrac{a}{\sin A} = 2R$

Similarly, by considering angles B and C, we can prove that $\dfrac{b}{\sin B} = 2R$ and $\dfrac{c}{\sin C} = 2R$ respectively.

Thus, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R.$

(i) The Law of Sines can be written as a collection of three equations

$$\frac{a}{b} = \frac{\sin A}{\sin B}; \qquad \frac{a}{c} = \frac{\sin A}{\sin C}; \qquad \frac{b}{c} = \frac{\sin B}{\sin C};$$

(ii) The Law of Sines says that the sides of a triangle are proportional to the sines of their opposite angles.

(iii) Using the Law of Sines, it is impossible to find the solution to a triangle given two sides and the included angle.

(iv) An interesting geometric consequence of the Law of Sines is that the largest side of any triangle is opposite to the largest angle. (Prove)

Napier's Formula

Theorem 3.2: In $\triangle ABC$, we have

(i) $\tan \dfrac{A-B}{2} = \dfrac{a-b}{a+b} \cot \dfrac{C}{2}$

(ii) $\tan \dfrac{B-C}{2} = \dfrac{b-c}{b+c} \cot \dfrac{A}{2}$

(iii) $\tan \dfrac{C-A}{2} = \dfrac{c-a}{c+a} \cot \dfrac{B}{2}$